

ELECTROCHEMISTRY

Electrochemistry is the branch of physical chemistry concerned with the relationship between electrical potential, as a measurable and quantitative phenomenon, and identifiable chemical changes, with either • electrical potential as an outcome of a particular chemical change, or vice versa.

The History and Timeline of Events

16th Century The Beginning:-

- 1550's William Gilbert became known as "the father of Magnetism". He made the first clear distinction between magnetism and what was then called the "amber effect".

17th Century static electricity:-

- 1663 Otto von Guericke created the first electrostatic generator producing static electricity by applying friction. Von Guericke used his generator to show that like charges repelled each other.



18th Century Birth of Electrochemistry:-

- 1709 Francis Hauksbee created the first gas-discharge lamp.
- Mid 18th century Charles Francois de Casterney Du Fay announced the 2-fluid theory of electricity.
- opposing Benjamin Franklin's 1-fluid theory.
- 1781 Charles-Augustin de Coulomb developed the law of electrostatic attraction.



- 1791 Luigi Galvani marked the birth of electrochemistry bridging muscular contraction & electricity in an essay "commentary" on the Effect of Electricity on Muscular Motion.

Alessandro Volta disputed this theory. Later his name was used for the unit of electrical potential, the volt.

19th Century The Rise of electrochemistry:-

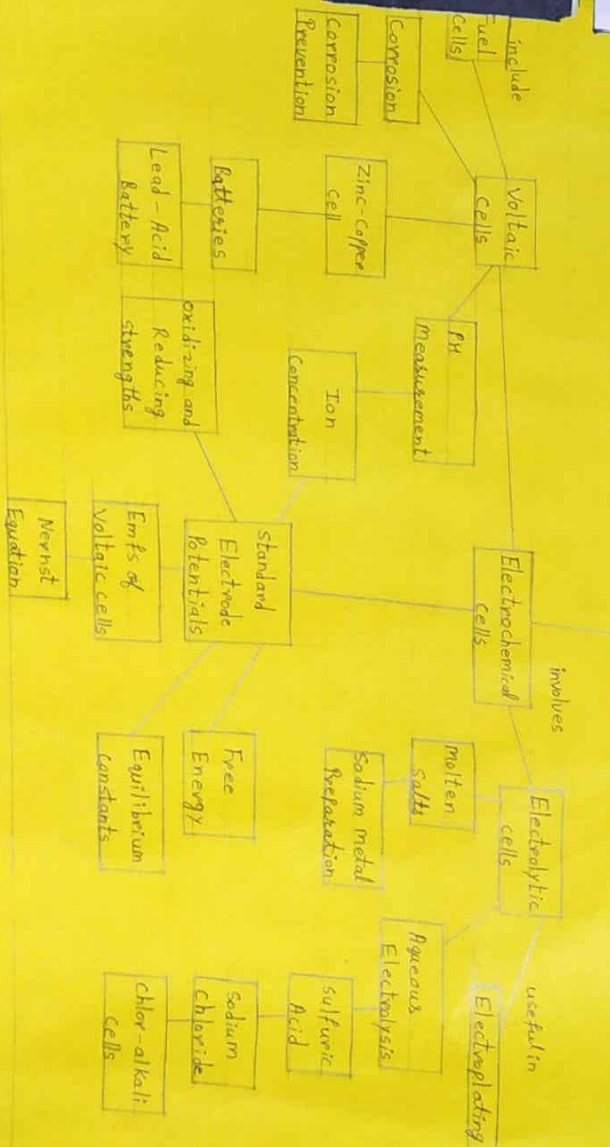
- 1800 Nicholson & Ritter succeeded in separating water into hydrogen & oxygen by electrolysis and later Ritter discovered the process of electroplating.
- 1801 William Cruickshank designed the first electrolytic.
- 1809 Goettinger developed the first telegraph.
- 1820 Michael Faraday contributed greatly to the fields of electromagnetics and electrochemistry.
- 1833 Wm. Grove produced the first fuel cell.
- 1836 Hestruitt & Hall won the race for the production of aluminum.
- 1836 Wilhelm Ostwald experimented with catalytic reactions and is known for contributions to electrochemistry.



20th Century THE Advent of Electrochemical Societies: 21st Century and Beyond

- 1902 the electrochemical society (ECS) was founded.
- 1903 Millikan began experiments to determine the electric charge carried by a single electron and in 1910 he obtained precise electron result with his famous oil-drop experiment.
- 1912 the "galvanograph" was born as Heyrovsky recorded the current voltage shortly thereafter with his Japanese colleague Masao Shikata, he constructed the first instrument for the automatic recording of polarographic curves, which became world famous later polarograph.
- 1949 the International Society of Electrochemistry (ISE) was founded.
- 1991 Commercial introduction of the lithium ion battery.
- 2003 US President George Bush proposed the Hydrogen Fuel Initiative (HFI) to further develop hydrogen fuel cells and infrastructure technology/fuel to accelerate the commercial introduction of fuel cell vehicles.
- 2005 wireless energy transfer - wireless - the ability to provide electrical energy to remote objects without wires.
- 2007 solid state micrometer-scale electric double-layer capacitors based on advanced superionic conductors had been for low-voltage electronics such as blue, sub-voltage electronics such as monoelectronics and related technology.
- 2007 The non-aqueous battery, a lithium ion battery, was invented by a team led by Dr. Y. Cui.

Electrochemistry



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